Module 1: The foundation

Cloud setup & access

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Prerequisites

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- **Providers:** DigitalOcean, AWS EC2, Linode, or Raspberry Pi.

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Learning objectives

- Deploy a server running **Ubuntu 24.04 LTS**.
- Generate and configure **SSH Keys** (Ed25519).
- Understand the **"Lock & Key"** security model.
- **Security:** Disable password authentication completely.

How SSH keys work

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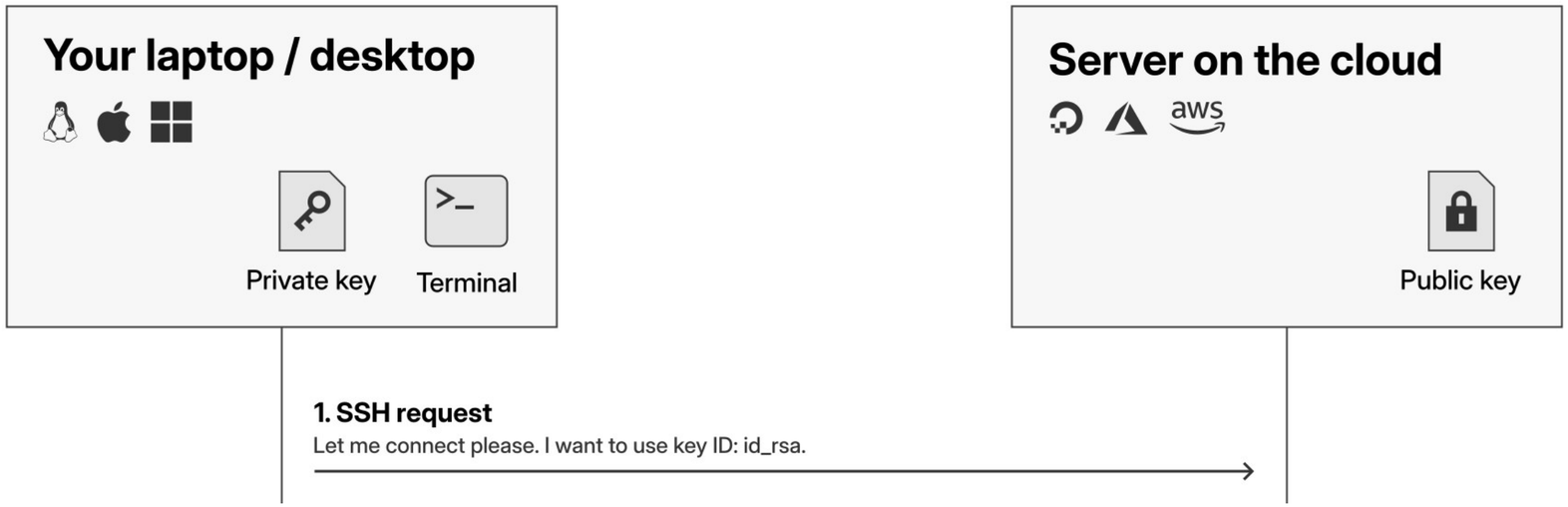

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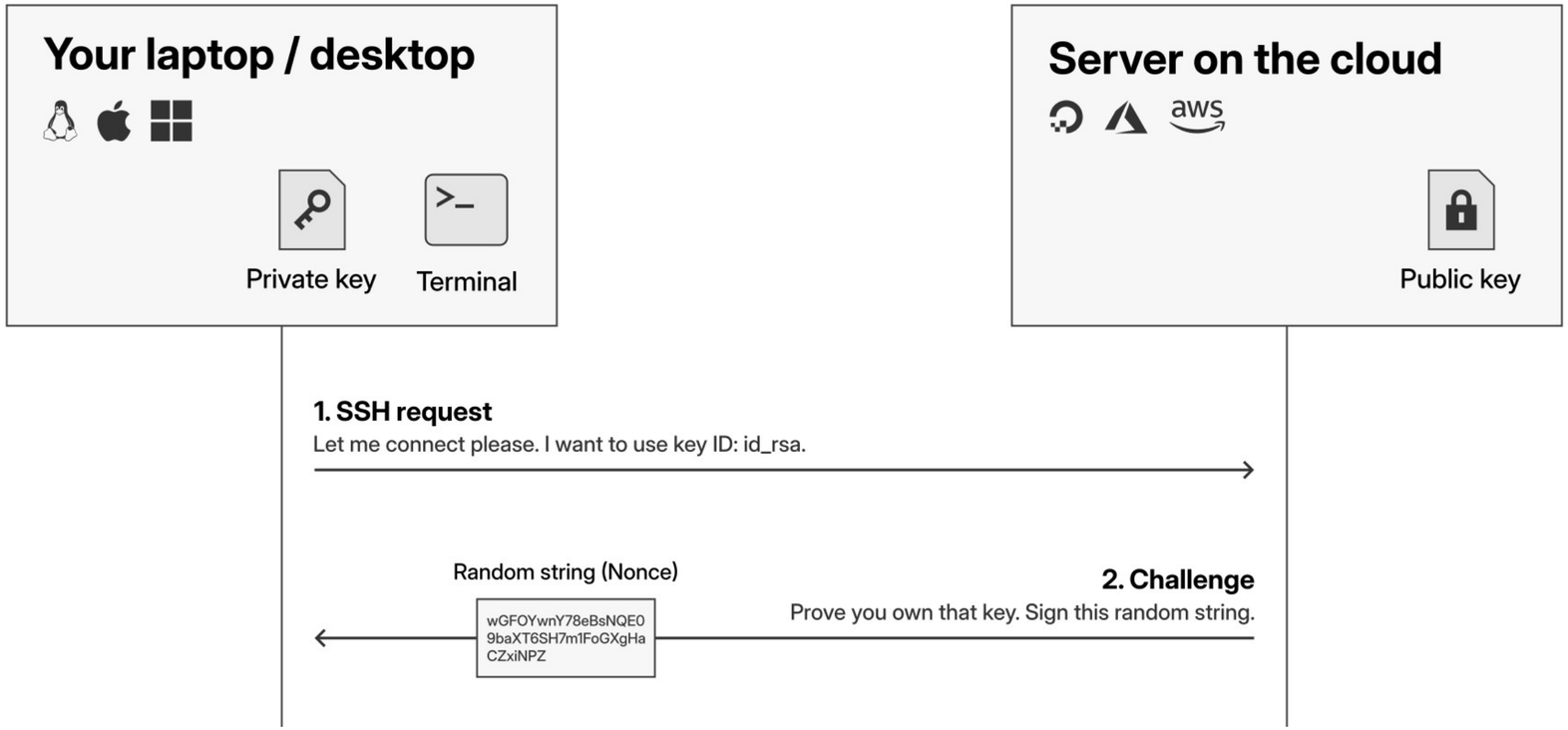

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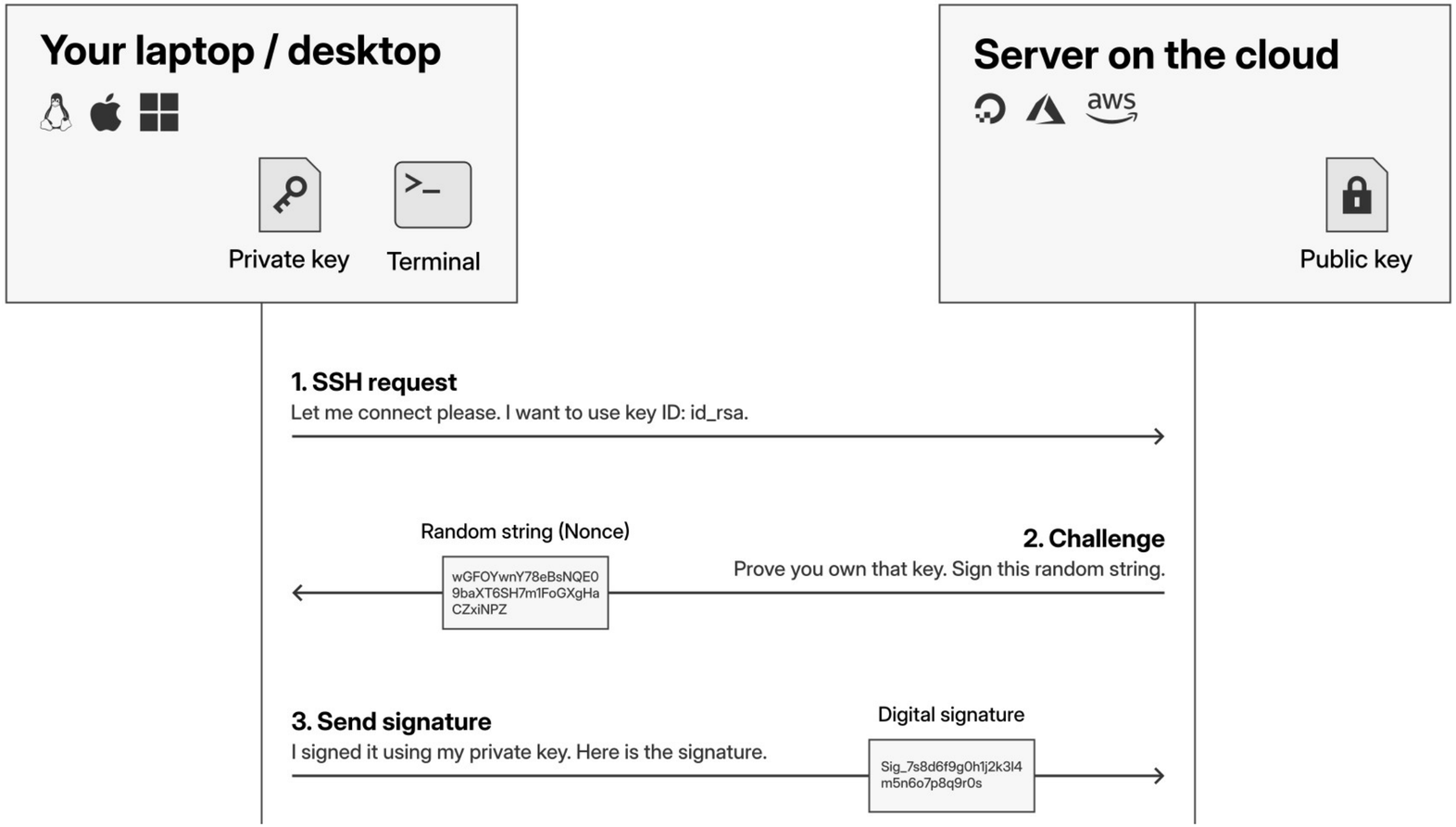

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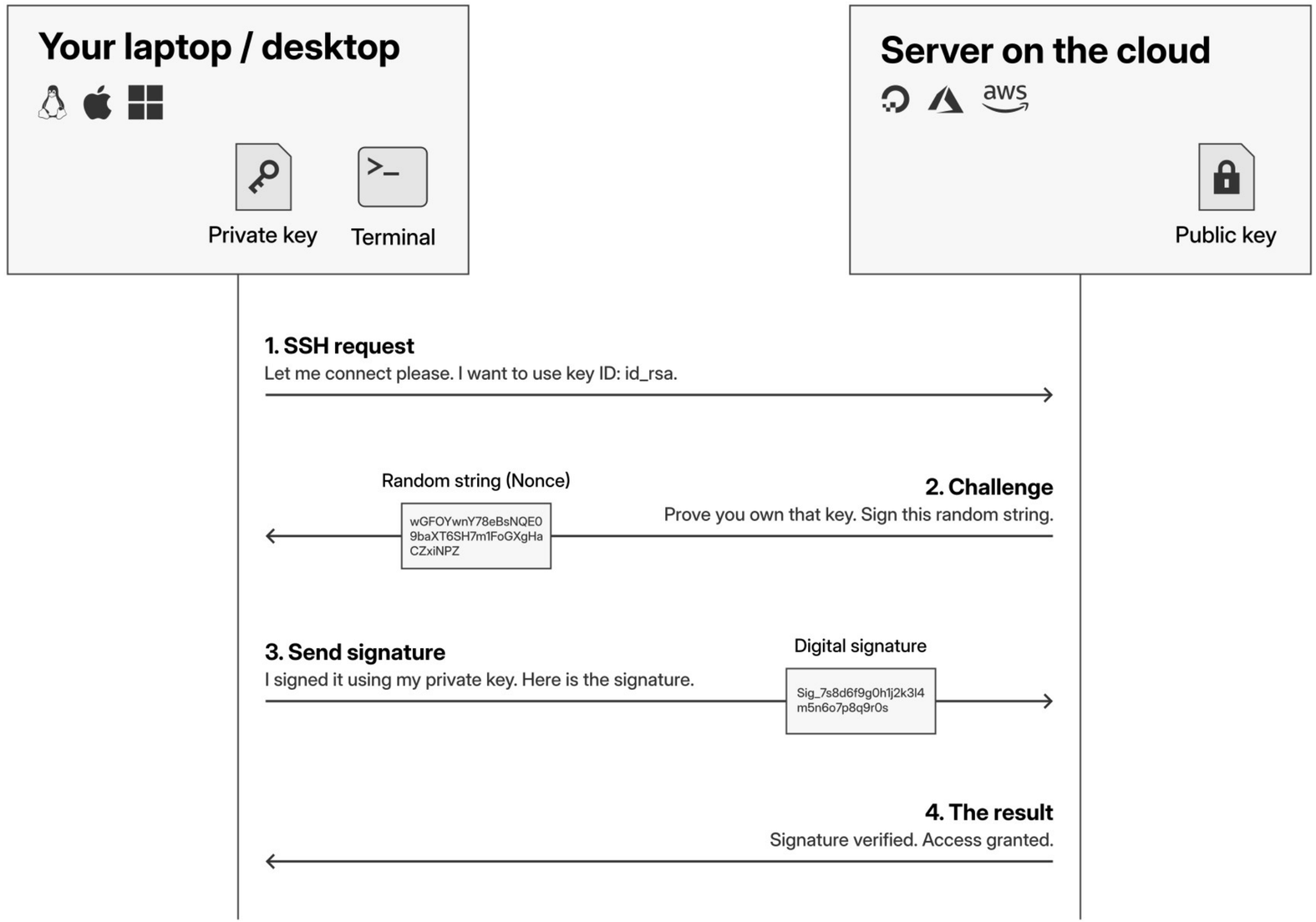

What is next?

- **Current status:** You now have a clean, secure Ubuntu server.
- **The risk:** The server is still exposed to the open internet.
- **Next steps:**
  - **Firewall configuration:** Setting up UFW (Uncomplicated Firewall).
  - **Brute force protection:** Installing Fail2Ban to block attackers.
  - **Emergency access:** How to use the recovery console.

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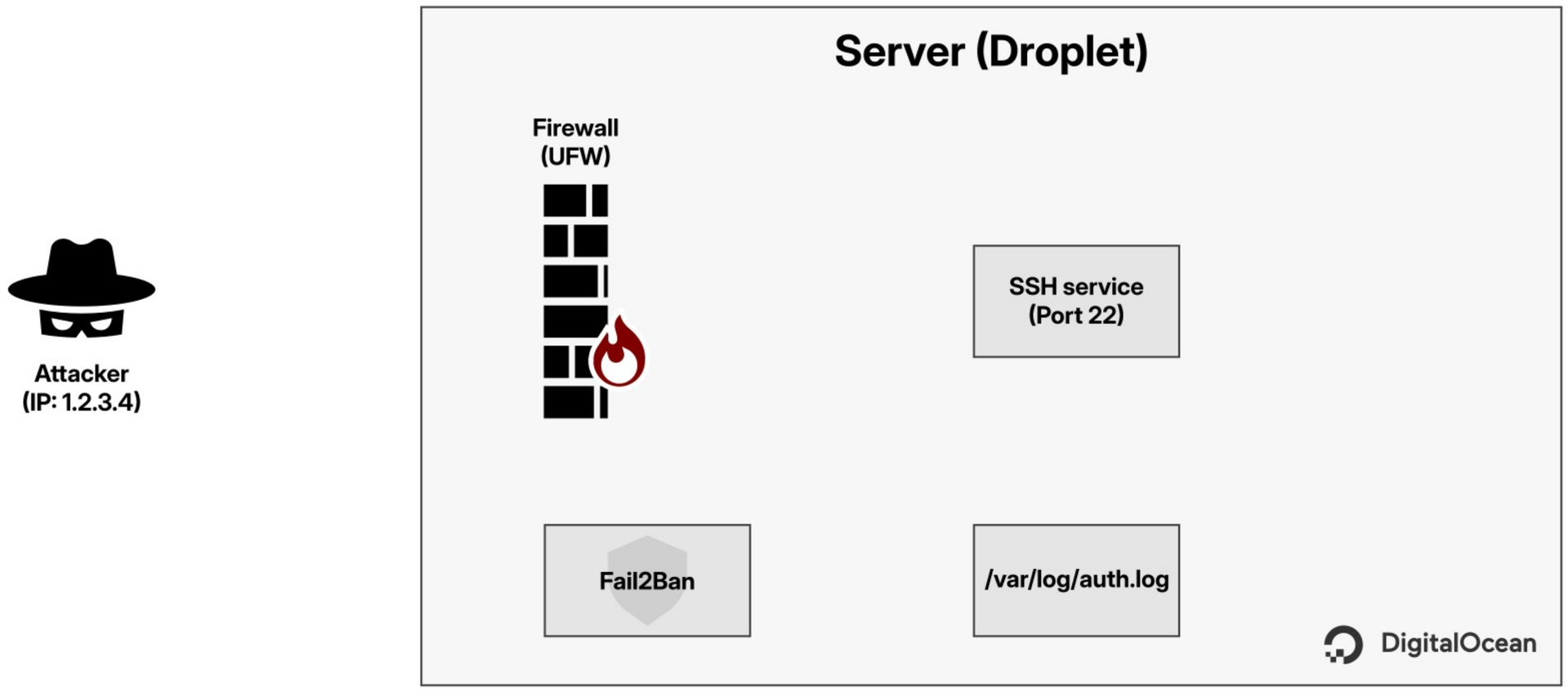
The firewall strategy

The firewall strategy

- **Problem:** Default Linux servers accept traffic on all ports.
- **Solution:** Use UFW to block everything by default.
- **Automation:** Use Fail2Ban to block bots instantly.
- **Safety net:** Using the DigitalOcean Recovery Console.

How Fail2Ban works



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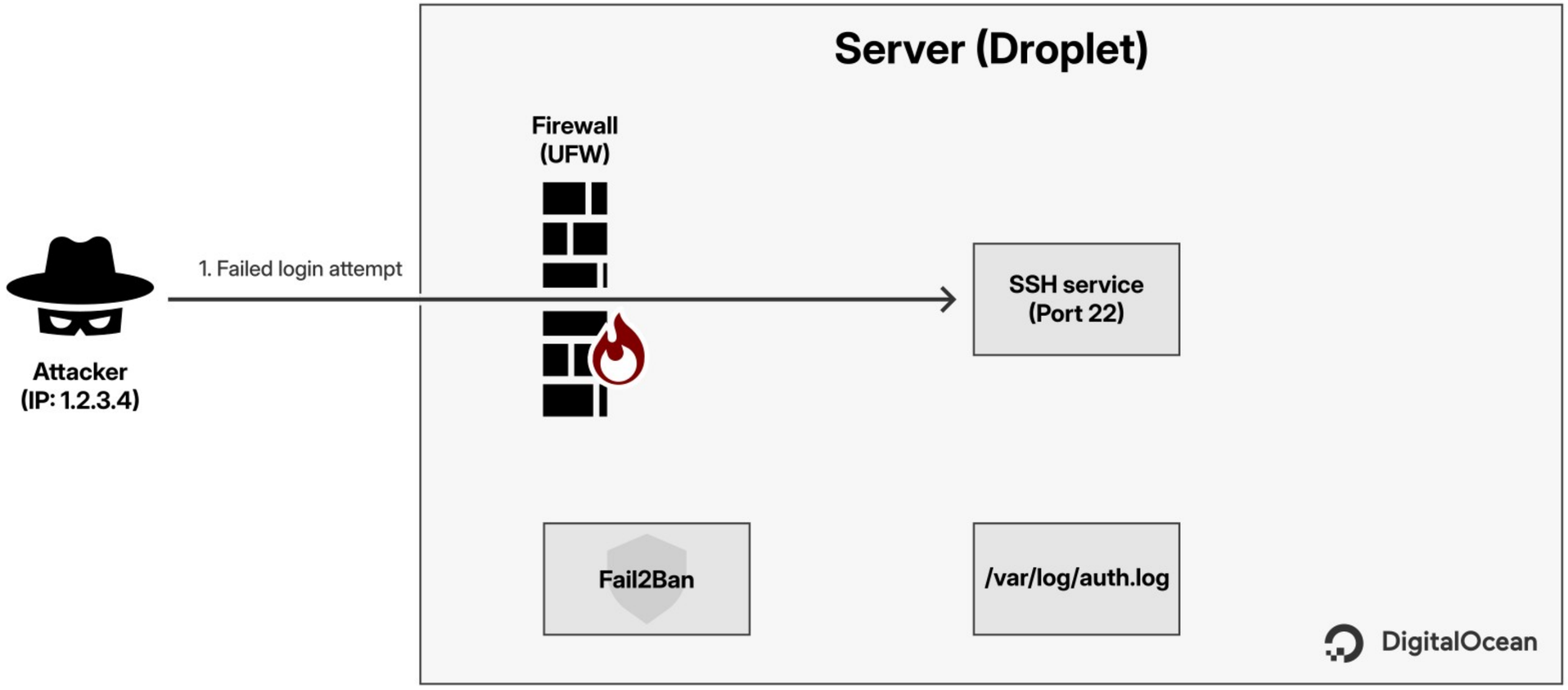

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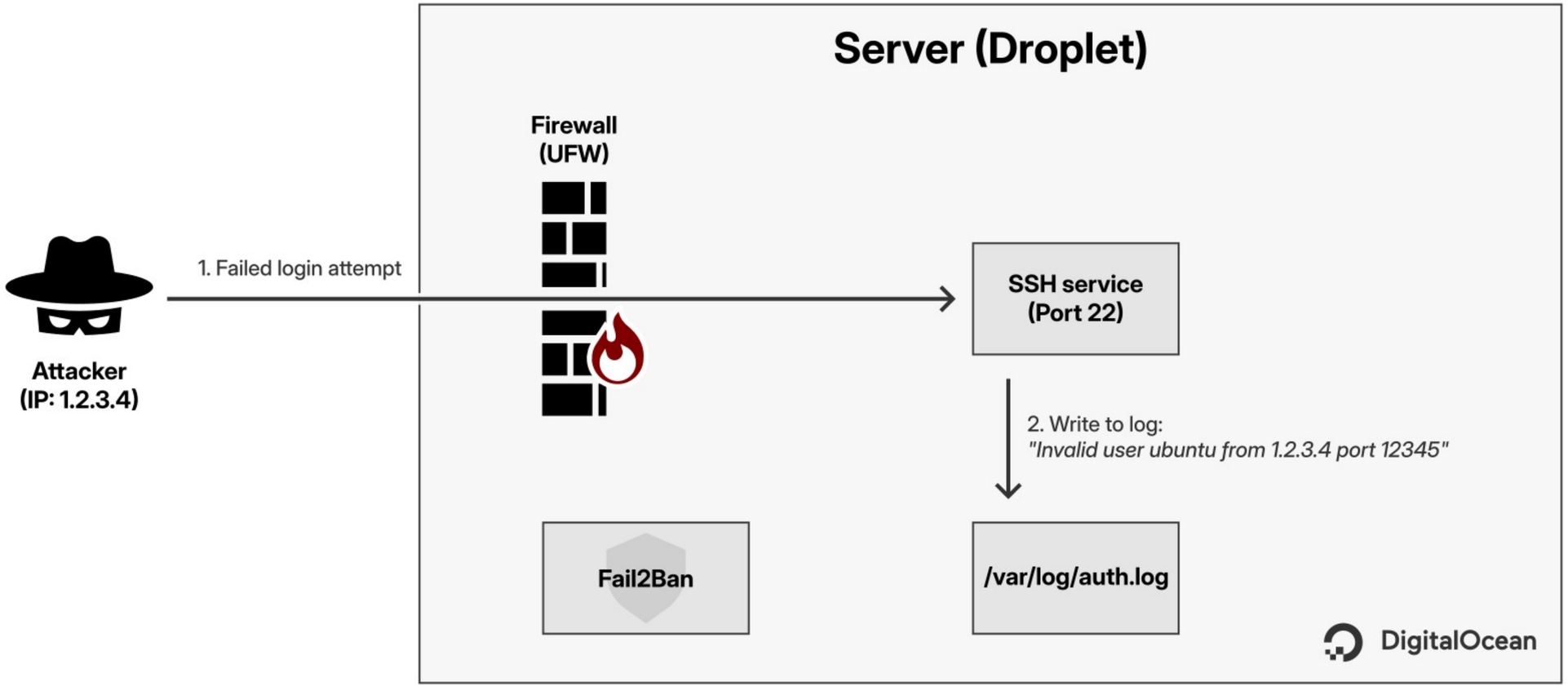

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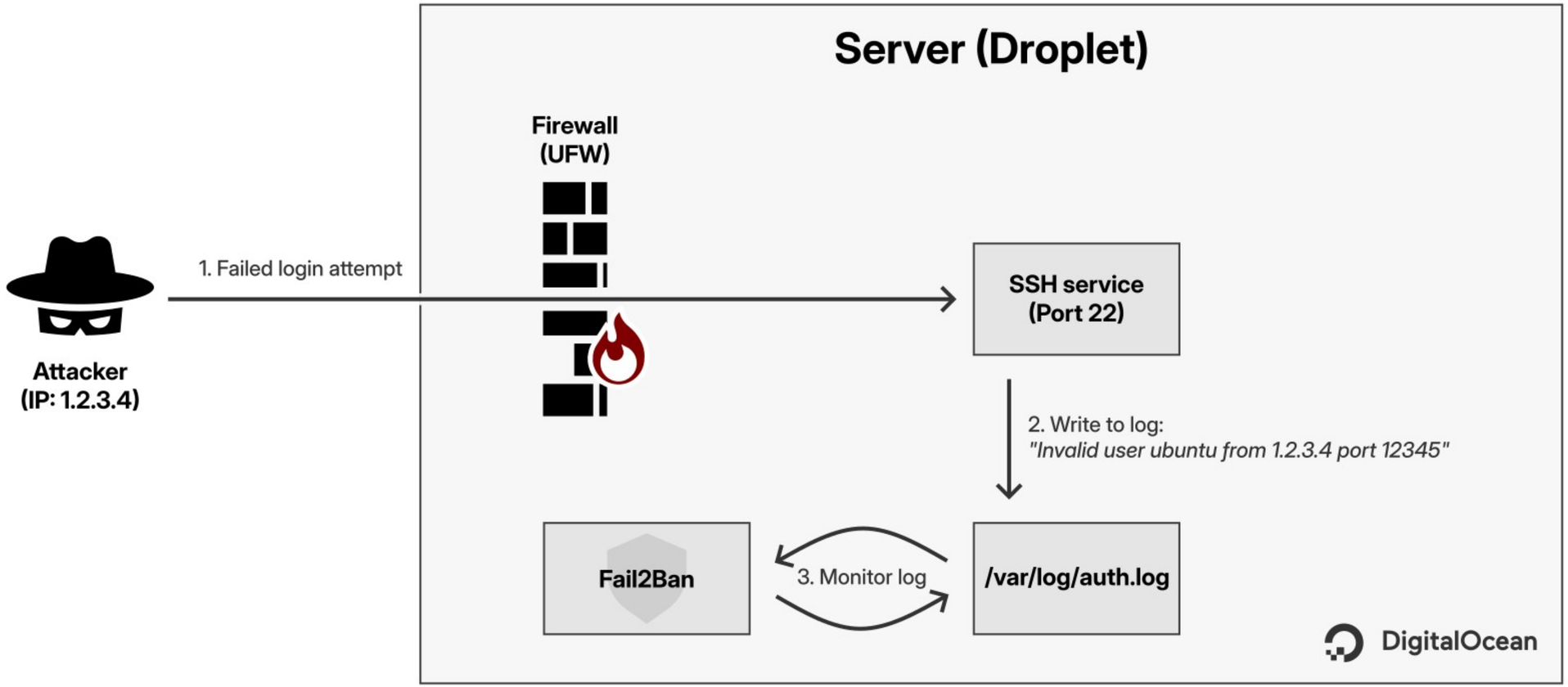

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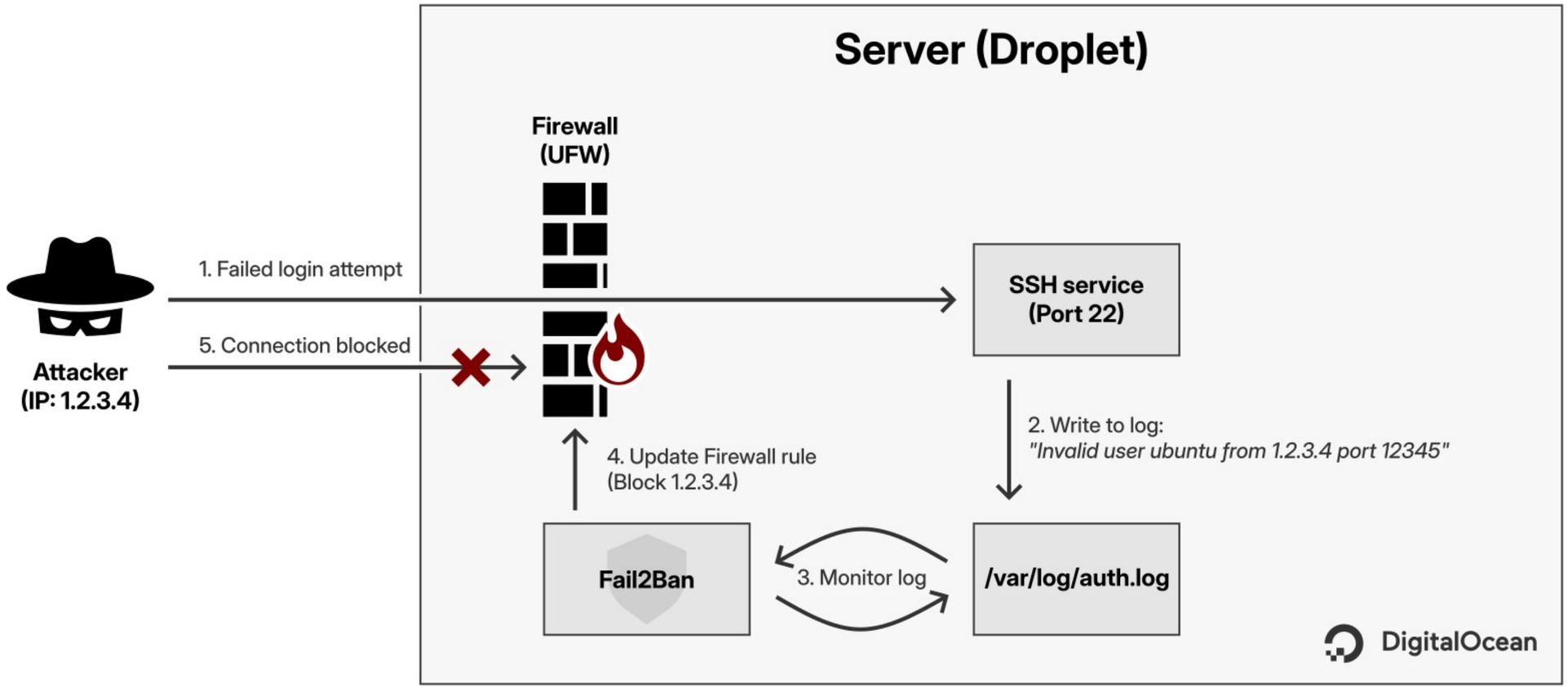

Module 2: The application runtime

Manual deployment

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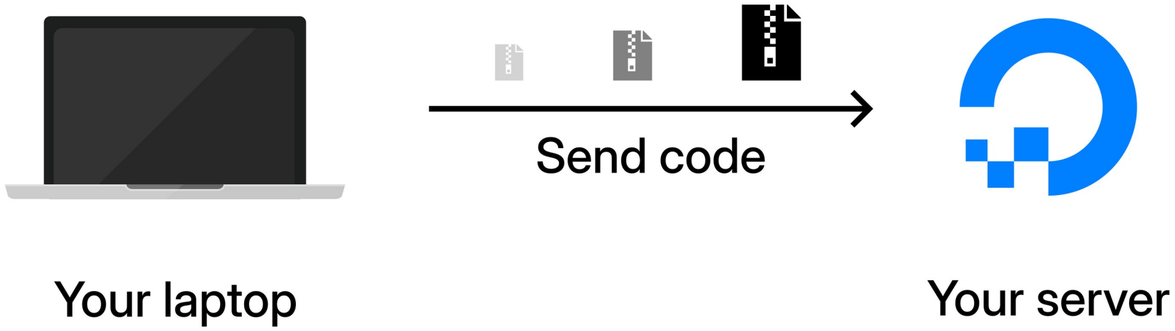
- Move code using **scp** and organize permissions.



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- Install **Python** (Backend) and **Node.js** (Frontend).
- Fix build errors using **Swap files**.
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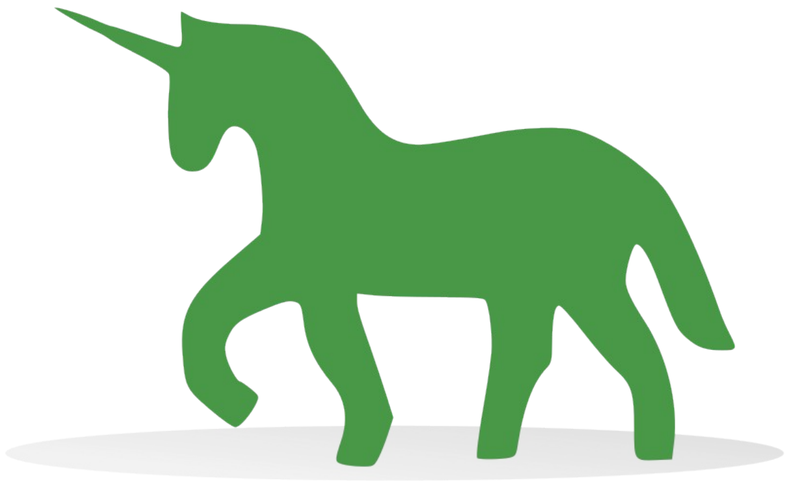
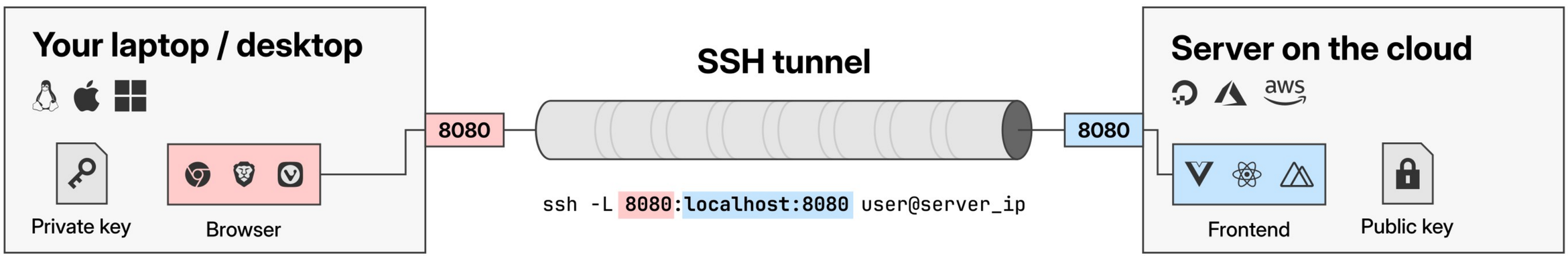
- Configure **Gunicorn** and **Supervisor**.



SSH tunneling



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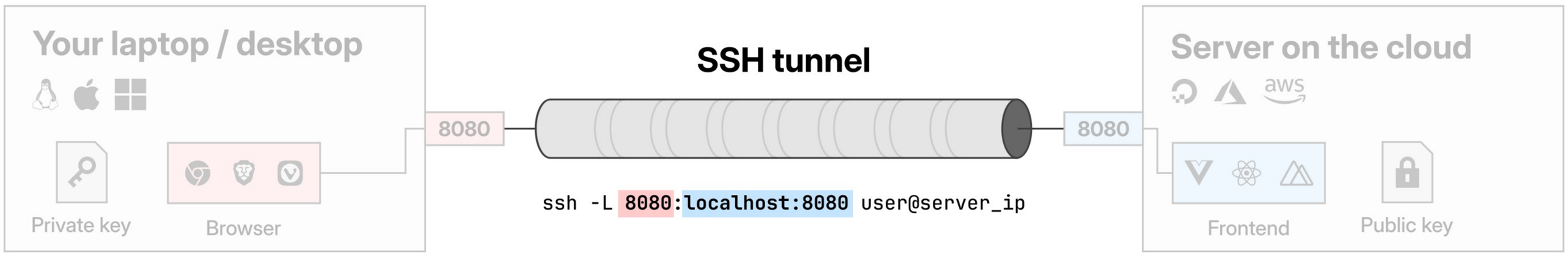

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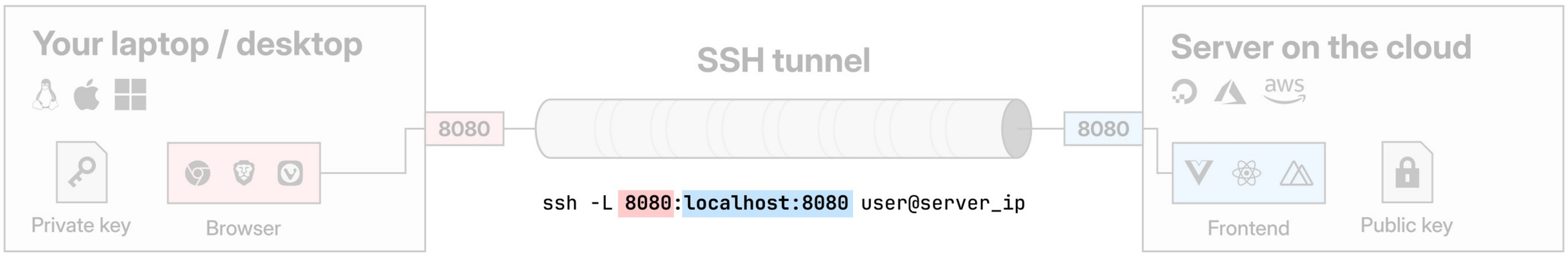

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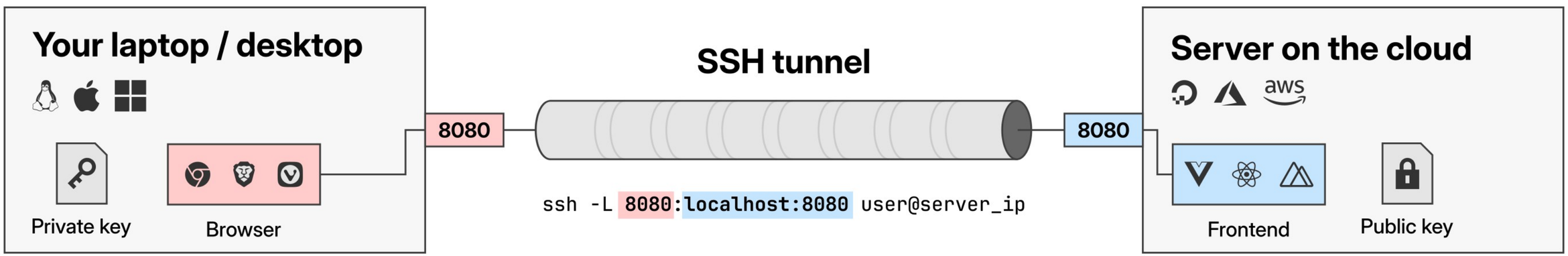
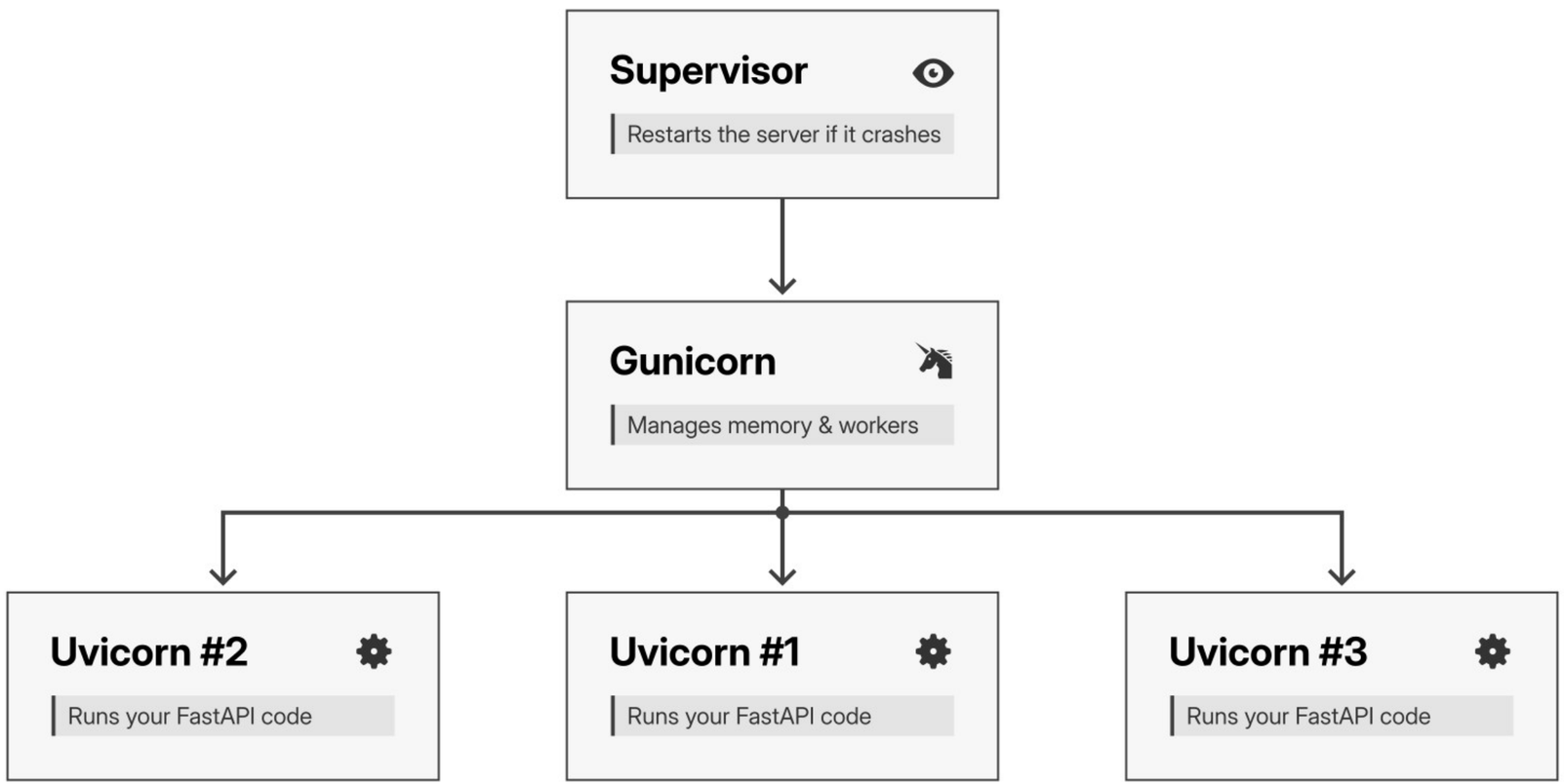


Keeping the app alive



Module 2: The application runtime

Reverse proxy & security headers

Reverse proxy & Security

- Sync frontend and backend URLs.
- Install and configure **Nginx**.

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Reverse proxy & Security

- Sync frontend and backend URLs.
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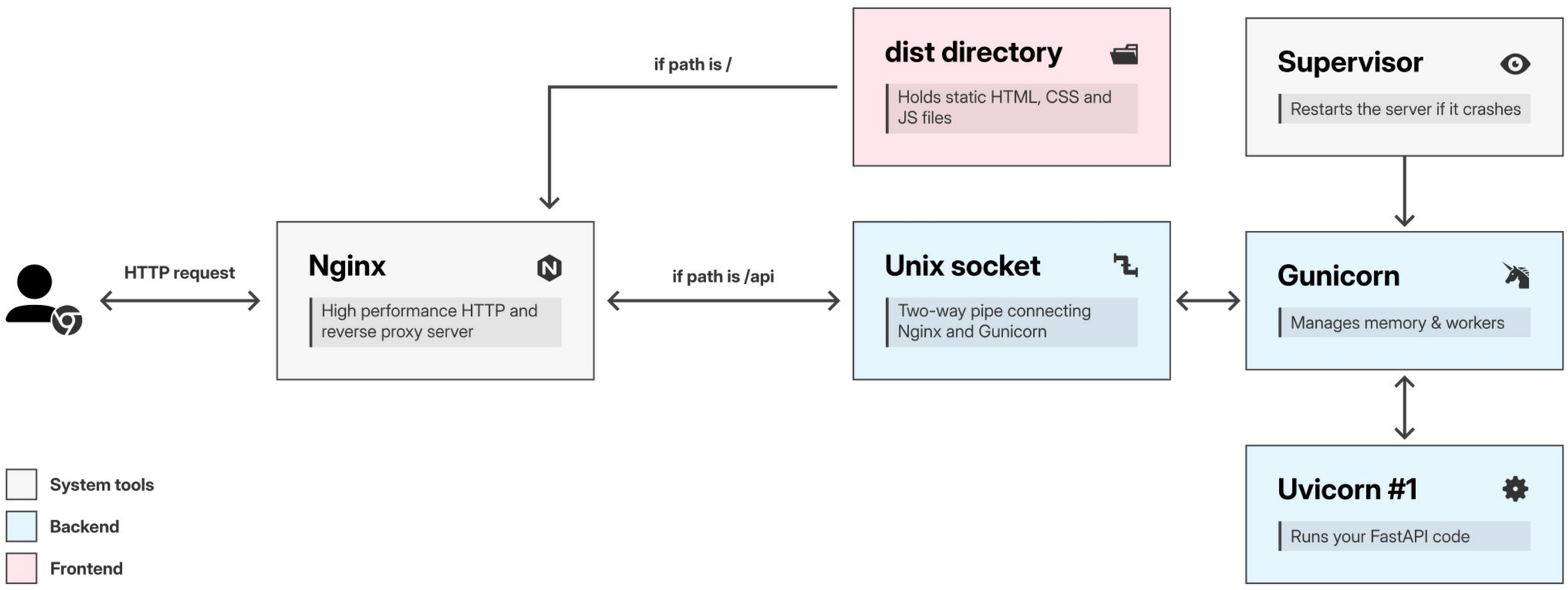
- Understand and fix the **502 Bad Gateway** error.
- Protect users with **HTTP Security Headers (CSP)**.

Nginx: The traffic cop